

# Embedded Triangular Graphs

T.H. Ku, J.S. Lin, Y.L. Wang and R.C.T. Lee

Department of Computer Science and Information Engineering

National Chi Nan University, Nantou, Taiwan, ROC

thku@alg.csie.ncnu.edu.tw, jslin2003@mail.stut.edu.tw, yuelwang@ncnu.edu.tw,  
retlee@ncnu.edu.tw

## Abstract

*In this paper, we define a special planar graph that we named embedded triangular graphs. We shall discuss some properties in the embedded triangular graph. We prove that the embedded triangular graph is an edge pancyclic graph and a panconnected graph. However, it is not a geodesic-pancyclic graph because there is a counterexample in it.*

## 1 Introduction

There are a wide variety of researches on graph theory. Many of them discuss the properties of graphs. Some important properties of graphs include pancyclicity, panconnected and Hamiltonian cycle, etc. Many kinds of graphs have been proved that they have the pancyclicity property [2,3,5,7,8] and many kinds of graphs also have been proved that they have the panconnected property [1]. But there are still many graphs that there are fewer discussions on their properties. In this paper, we shall define a special planar graph and try to find some properties in it. Finally, we prove that this special planar graph has the pancyclicity and panconnected properties.

## 2 Embedded triangular graphs

In the following, we shall define a planar graph called an embedded triangular graph which is constructed as follows: First, we start with an arbitrary triangle in the plane which is called a 1-triangle whose vertices are denoted as  $v_1$ ,  $v_2$  and  $v_3$ , and the edges of 1-triangle are called 1-triangle edges and also called 1-edges. Second, namely step 2, we select any point, denoted as  $v_4$ , inside the triangle and connect  $v_4$  with the three vertices of 1-triangle, as shown in Fig. 1. All of the triangles created in this step are called 2-triangles and all of the edges created in this step are called 2-edges. The edges of 2-triangles are called 2-triangle edges.

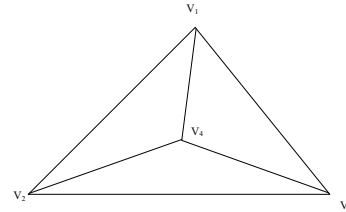
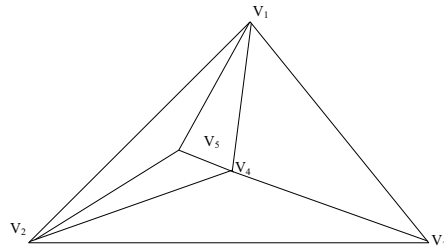
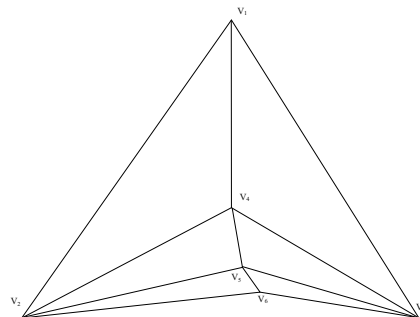


Figure 1: 2-triangles.

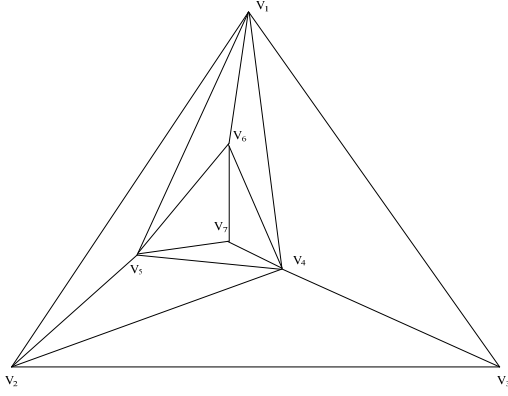
In the following steps, we select any triangle created in Step  $i$ , select a point  $v_{i+3}$  inside the triangle and connect  $v_{i+3}$  to the three vertices of the  $i$ -triangle. The resulting graph is an embedded triangular graph. Note that a triangle created in step  $i$  is called an  $i$ -triangle and all edges created in this step are called  $i$ -edges. The edges of  $i$ -triangles are called  $i$ -triangle edges. Fig. 2 shows several examples of embedded triangular graphs.



(a) An embedded triangular graph with 5 vertices



(b) An embedded triangular graph with 6 vertices



(c) An embedded triangular graph with 7 vertices

Figure 2 Examples of embedded triangular graphs.

### 3. The Edge Pancyclic Property of Embedded Triangles.

#### Definition of $k$ -cycle edge:

Given a graph  $G = (V, E)$ , an edge is a  $k$ -cycle edge if and only if it is contained in a cycle of length  $k$  [6].

#### Definition of Edge Pan-cyclic:

A graph  $G$  with  $n$  vertices is edge pancyclic if all edges in  $G$  are  $k$ -cycle, for all  $k$ , where  $3 \leq k \leq n$  [5].

**Theorem 3.1:** Every embedded triangular graph is an edge pan-cyclic graph.

Proof:

We prove this theorem by mathematical induction. First, let us consider an embedded graph with 3 vertices which is the basis. This graph is a 1-triangle embedded triangular graph. It is easy to prove that every 1-edge is a 3-cycle edge. Therefore, the embedded triangular graph with 3 vertices is an edge pancyclic graph.

Next, let us assume that the embedded triangular graph with  $n$  vertices is an edge pancyclic graph. That is, all edges in the graph are  $k$ -cycle, where  $3 \leq k \leq n$ .

Now, let us consider the embedded graph with  $n+1$  vertices. By the definition of embedded triangular graph, vertex  $v_{n+1}$  is inside a  $(n-2)$ -triangle and is connected to three vertices of an  $(n-2)$ -triangle. These newly constructed triangles are  $(n-1)$ -triangles. We want to show that all edges in the embedded triangular graph with  $n+1$  vertices are  $k$ -cycle, where  $3 \leq k \leq n+1$ . We need to show (1) For every

$j$ ,  $1 \leq j \leq n-2$ , a  $j$ -edge is  $k$ -cycle for  $3 \leq k \leq n+1$ . (2) The  $(n-1)$ -edges are  $k$ -cycle for  $3 \leq k \leq n+1$ .

(1) By the inductive hypothesis, for every  $j$ ,  $1 \leq j \leq n-2$ , a  $j$ -edge is  $k$ -cycle for  $3 \leq k \leq n$ . We now show that such a  $j$ -edge is also  $(n+1)$ -cycle. We have to discuss two cases:

Case 1: Assume that  $v_{n+1}$  is inside a certain  $(n-2)$ -triangle. Consider any edge of this triangle. This edge must be a  $j$ -edge and connected to  $v_{n+1}$ . Now, without loss of generality, we assume that the embedded triangular graph with  $(n+1)$  vertices starts with  $\Delta v_1 v_2 v_3$ , i.e., a triangle with three vertices,  $v_1$ ,  $v_2$  and  $v_3$ . There exists a graph obtained by deleting, say  $v_1$ , is also an embedded triangular graph with  $n$  vertices. By the definition of embedded triangular graphs, the  $j$ -edge mentioned above must be in this embedded triangular graph with  $n$  vertices. By the inductive hypothesis, this  $j$ -edge is  $k$ -cycle for  $3 \leq k \leq n$ . Thus there exists a cycle  $C$  of length  $n$  containing the  $j$ -edge. Since  $C$  is of length  $n$ , it must contain every vertex of the graph. Thus,  $C$  contains vertex  $v_2$  and consequently an edge connected to  $v_2$ . Without loss of generality, we assume this edge is  $\overline{v_2 v_3}$ . Thus we can find a cycle of length  $(n+1)$  containing the  $j$ -edge by replacing  $\overline{v_2 v_3}$  by  $\overline{v_2 v_1}$  and  $\overline{v_1 v_3}$  in the embedded triangular graph with  $(n+1)$  vertices starting with  $\Delta v_1 v_2 v_3$ .

Case 2: Now, we consider those  $j$ -edges which are not connected to  $v_{n+1}$ . Note that any cycle with  $n$  edges which contains the  $j$ -edge must contain an  $(n-2)$ -edge; otherwise, it cannot be a cycle with  $n$  edges in the embedded triangular graph with  $n$  vertices. Without loss of generality, we assume that this  $(n-2)$ -edge is  $\overline{v_{n-1} v_n}$ . Now, we can construct a cycle of length  $(n+1)$  by using two  $(n-1)$ -edges,  $\overline{v_{n-1} v_{n+1}}$  and  $\overline{v_{n+1} v_n}$ , to replace the  $(n-2)$ -edge,  $\overline{v_{n-1} v_n}$ . Thus, every  $j$ -edge is also  $(n+1)$ -cycle, for  $1 \leq j \leq n-2$ .

(2) Consider any  $(n-1)$ -edge  $\overline{v_p v_{n+1}}$ . This

edge must be adjacent to  $\overline{v_p v_n}$  and  $\overline{v_{n+1} v_n}$ , where  $\overline{v_p v_n}$  is an  $(n-2)$ -edge and  $\overline{v_{n+1} v_n}$  is an  $(n-1)$ -edge. By inductive hypothesis,  $\overline{v_p v_n}$  is  $k$ -cycle for  $3 \leq k \leq n$ . Thus by replacing  $\overline{v_p v_n}$  by  $\overline{v_p v_{n+1}}$  and  $\overline{v_{n+1} v_n}$ , we can now conclude that  $\overline{v_p v_{n+1}}$  is  $k$ -cycle for  $3 \leq k \leq n+1$ .

Therefore, we have that every embedded triangular graph is an edge pan-cyclic graph.

#### 4. The Panconnected Property of Embedded Triangles.

Given a graph  $G = (V, E)$ , the shortest distance between two vertices  $x$  and  $y$  in the graph  $G$  is denoted by  $d(x, y)$ .

##### Definition of Panconnected:

A graph  $G$  is panconnected if there exists a path joining any two different vertices  $x$  and  $y$  with length  $l$ , for all  $l$ , where  $d(x, y) \leq l \leq |V(G)| - 1$ . [8]

Now we want to show an observation of embedded triangular graphs. Let  $G$  be an embedded triangular graph with  $n$  vertices. We assume that there exists a path of length  $(n-1)$  between two vertices  $v_i$  and  $v_j$ , where  $1 \leq i \neq j \leq n$ . If the path contains some two  $(n-2)$ -triangle edges, when a new vertex is constructed inside an  $(n-2)$ -triangle of the embedded triangular graph, we can easily find a path of length  $n$  between  $v_i$  and  $v_j$ . The new path of length  $n$  between  $v_i$  and  $v_j$  would still contain some two  $(n-1)$ -triangle edges. Now we shall show you what the some two  $(n-2)$ -triangle edges are. There are six kinds of combinations of the some two  $(n-2)$ -triangle edges fitting with the above discussing. Fig. 3 shows these six kinds of combinations of the some two 3-triangle edges in an embedded triangular graph with 5 vertices.

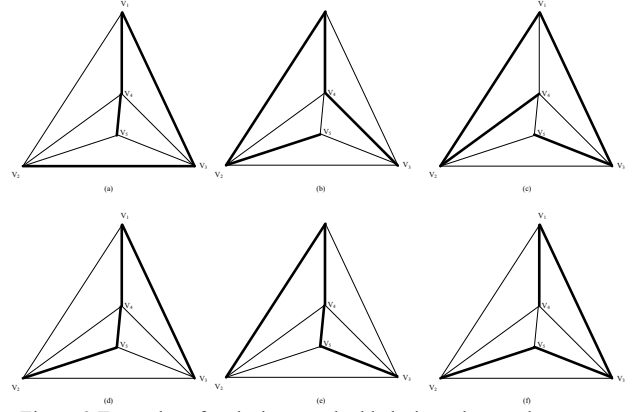


Figure 3 Examples of paths in an embedded triangular graph with 5 vertices

Fig. 3 shows that each path contains two  $(n-2)$ -triangle edges. For example, Fig. 3(a) shows a path of length 4 between  $v_5$  and  $v_2$ . This path contains two 3-triangle edges which are  $\overline{v_2 v_3}$  and  $\overline{v_5 v_4}$ . By the definition of embedded triangular graphs, when we construct a new vertex  $v_6$ , it must be in a 3-triangle and be connected to the vertices of the 3-triangle. Each of  $\overline{v_2 v_3}$  and  $\overline{v_5 v_4}$  is an edge of a 3-triangle. Thus we can find a larger path between  $v_5$  and  $v_2$  by replacing one of the two 3-triangle edges and the new path shall contain some two 4-triangle edges. Lemma 4.1 shall show you that there exists a path of length  $n-1$  between each two vertices  $v_i$  and  $v_j$  which contains the two  $(n-2)$ -triangle edges in an embedded triangular graph with  $n$  vertices. In Lemma 4.1 we shortly write two  $(n-2)$ -triangle edges to represent the some two  $(n-2)$ -triangle edges.

**Lemma 4.1:** In an embedded triangular graph with  $n$  vertices, for each two vertices  $v_i$  and  $v_j$ , there exists a path of length  $(n-1)$  between them, which contains two  $(n-2)$ -triangle edges, where  $n \geq 3$ .

Proof:

It is easy to prove that Lemma 4.1 holds in an embedded triangular graph with 3 vertices.

Next, we assume that Lemma 4.1 is true in an embedded triangular graph with  $n-1$  vertices. By the inductive hypothesis, there exists a path of length  $n-2$  between any two difference vertices  $v_i$  and  $v_j$  which contains two  $(n-3)$ -triangle edges in an embedded triangular graph with  $n-1$  vertices, where  $1 \leq i \neq j \leq n-1$ . Now, let us consider the embedded graph with  $n$  vertices.

By the definition of embedded triangular graph, vertex  $v_n$  is inside a  $(n-3)$ -triangle and is connected to three vertices of an  $(n-3)$ -triangle. These newly constructed triangles are  $(n-2)$ -triangles. We want to show that there exists a path of length  $n-1$  between any two difference vertices  $v_i$  and  $v_j$  which contains two  $(n-2)$ -triangle edges in an embedded triangular graph with  $n$  vertices, where  $1 \leq i \neq j \leq n$ . We need to show (1) There exists a path of length  $n-1$  between any two difference vertices  $v_i$  and  $v_j$  which contains two  $(n-2)$ -triangle edges in an embedded triangular graph with  $n$  vertices,  $1 \leq i \neq j \leq n-1$ . (2) There exists a path of length  $n-1$  between  $v_k$  and  $v_n$  which contains two  $(n-2)$ -triangle edges in an embedded triangular graph with  $n$  vertices, where  $1 \leq k \leq n-1$ .

(1) By the inductive hypothesis, there exists a path of length  $n-2$  between any two difference vertices  $v_i$  and  $v_j$  which contains two  $(n-3)$ -triangle edges, where  $1 \leq i \neq j \leq n-1$ . We can easily find a path of length  $n-1$  between  $v_i$  and  $v_j$  by replacing a  $(n-3)$ -triangle edge by two  $(n-2)$ -edges. Thus there exists a path of length  $n-1$  between any two difference vertices  $v_i$  and  $v_j$  which contains two  $(n-2)$ -triangle edges, where  $1 \leq i \neq j \leq n-1$ .

(2) Without loss of generality, we assume that the embedded triangular graph with  $n$  vertices start with  $\Delta v_1 v_2 v_3$ . There exists a graph obtained by deleting, say  $v_1$ , is also an embedded triangular graph with  $n-1$  vertices and this embedded triangular graph of  $n-1$  vertices must be an embedded triangular graph which starts with  $\Delta v_2 v_3 v_4$ . By the inductive hypothesis, there exists a path of length  $n-2$  between  $v_k$  and  $v_n$  which contains two  $(n-2)$ -triangle edges in the embedded triangular graph  $\Delta v_2 v_3 v_4$ , where  $2 \leq k \leq n-1$ . By the definition of embedded triangular graphs, one of  $v_2$ ,  $v_3$  and  $v_4$  is just connected to 3 vertices in  $\Delta v_2 v_3 v_4$ . Without loss of generality, we assume this vertex is  $v_2$ . Since a path of length  $n-2$  between  $v_k$  and  $v_n$  which contains two  $(n-2)$ -triangle edges, where  $3 \leq k \leq n-1$ , must contain  $v_2$ . Thus by the definition of embedded triangular graphs, the path of length  $n-2$  between  $v_k$  and  $v_n$  which contains two  $(n-2)$ -triangle edges, where

$3 \leq k \leq n-1$ , must contain a 1-edge of the embedded triangular graph  $\Delta v_2 v_3 v_4$ . This 1-edge of the embedded triangular graph  $\Delta v_2 v_3 v_4$  may be  $\overline{v_2 v_3}$  or  $\overline{v_2 v_4}$ . Thus we can find a path of length  $(n-1)$  between  $v_k$  and  $v_n$  which contains two  $(n-2)$ -triangle edges by replacing the 1-edge by  $\overline{v_2 v_1}$  and  $\overline{v_1 v_3}$  (or  $\overline{v_2 v_1}$  and  $\overline{v_1 v_4}$ ), where  $3 \leq k \leq n-1$ . Since there exists a path of length  $n-2$  between  $v_n$  and  $v_2$  which contains two  $(n-2)$ -triangle edges and  $v_1$  is adjacent to  $v_2$ , we can find a path of length  $n-1$  between  $v_n$  and  $v_1$  by concatenating  $v_1$  and the path between  $v_n$  and  $v_2$ . Finally we shall show you that there exists a path of length  $(n-1)$  between  $v_n$  and  $v_2$ . By the inductive hypothesis, there must exist a path of length  $(n-3)$  between  $v_n$  and  $v_3$  which contains two  $(n-2)$ -triangle edges and this path does not contain  $v_2$ . We can find a path of length  $(n-2)$  by concatenating  $v_2$  and the path of length  $(n-3)$  between  $v_n$  and  $v_3$  which contains two  $(n-2)$ -triangle edges. Finally we can find a path of length  $(n-1)$  between  $v_n$  and  $v_2$  which contains two  $(n-2)$ -triangle edges by replacing  $\overline{v_2 v_3}$  by  $\overline{v_2 v_1}$  and  $\overline{v_1 v_3}$ . This completes the proof.

**Theorem 4.2:** Every embedded triangular graph is a panconnected graph.

Proof:

First, let us consider an embedded triangular graphs with 3. In this embedded graph, the distance between every two vertices is 1. It is easy to prove that the graph is panconnected.

We assume that the embedded triangular graph with  $n$  vertices is a panconnected graph. This means that for every  $v_i$  and  $v_j$   $1 \leq i \neq j \leq n-1$ , there is a path between  $v_i$  and  $v_j$  with length  $l$  such that  $d(v_{n+1}, v_k) \leq l \leq n-1$ . Now, let us consider the embedded graph with  $n+1$  vertices. We need to show (1) For every two vertices  $v_i$  and  $v_j$  in this graph, there also exists a path of length  $n$ , where  $1 \leq i \neq j \leq n$ . (2) For  $v_{n+1}$  and  $v_k$ , there exists a path of length  $l$ , where  $d(v_{n+1}, v_k) \leq l \leq n$  and  $1 \leq k \leq n$ .

By assumption, there exists a path joining any

two different vertices  $v_i$  and  $v_j$  with length  $l$ , for all  $l$ , where  $d(v_i, v_j) \leq l \leq n-1$  and  $1 \leq i \neq j \leq n$ . Thus, now we need to show that there exists a path joining any two different vertices  $v_i$  and  $v_j$  with length  $n$ . By Lemma 4.1, for each two vertices  $v_i$  and  $v_j$ , there exists a path of length  $n-1$  including a  $(n-2)$ -triangle edge and we assume that this edge is  $\overline{v_m v_n}$ , where  $1 \leq i \neq j \leq n$  and  $m < n$ . Since the vertex  $v_{n+1}$  is connected to  $v_m$  and  $v_n$ , we can replace  $\overline{v_m v_n}$  by two edges  $\overline{v_m v_{n+1}}$  and  $\overline{v_{n+1} v_n}$ . Thus, there exists a path joining any two different vertices  $v_i$  and  $v_j$  with length  $l$ , for all  $l$ , where  $d(v_i, v_j) \leq l \leq n$  and  $1 \leq i \neq j \leq n$ .

By the definition of embedded triangular graphs,  $v_{n+1}$  is adjacent to three vertices of an  $(n-2)$ -triangle. Since  $v_{n+1}$  is connected to three vertices of an  $(n-2)$ -triangle, a path between  $v_{n+1}$  and  $v_k$  must pass one of the three vertices, say  $v_a$ , of the  $(n-2)$ -triangle, where  $1 \leq k \leq n$ . Since the embedded triangular graph with  $n$  vertices is a panconnected graph, there exists a path of length  $l$  between  $v_a$  and  $v_k$  in the embedded triangular graph, where  $d(v_a, v_k) \leq l \leq n-1$  and  $1 \leq k < n$ . Thus, a path between  $v_{n+1}$  and  $v_k$  of length  $l$ , where  $d(v_n, v_k) \leq l \leq n$ , can be found by concatenating edge  $\overline{v_n v_{n+1}}$  with the path between  $v_n$  and  $v_k$ . This completes the proof.

In Fig 3(a), consider  $v_5$ . We want to show that there exists a path of length 4 between  $v_5$  and  $v_1$ . Note that  $v_5$  is adjacent to  $v_3$ . There exists a path between  $v_1$  and  $v_3$ , as shown in Fig. 3(b) whose length is 3. By concatenating  $\overline{v_5 v_3}$  with this path, we obtain a path between  $v_5$  and  $v_1$  with length 4.

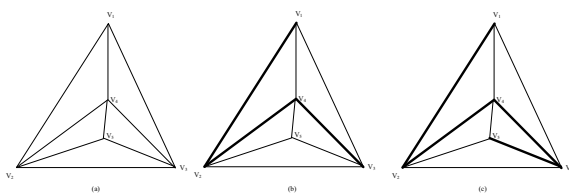


Figure 3 An illustrating example.

## 5. Conclusion

In this paper, we have shown that an embedded triangular graph has the edge pancyclic and panconnected properties.

## Reference:

- [1] Y. Alavi and J. E. Williamson, Panconnected graphs, *Studia Scientiarum Mathematicarum Hungarica* 10 (1975) 19–22.
- [2] J. A. Bondy, “Pancyclic graphs I,” *J. Combinat. Theory*, Vol. 11, pp. 80-84, 1971.
- [3] H. J. Broersma, A note on the minimum size of a vertex pancyclic graph, *Discrete Mathematics*, 164 (1997) 29-32.
- [4] H. C. Chan, J.M. Chang, Y. L. Wang and S.J. Horng, Geodesic-Pancyclic Graphs, *The 23rd Workshop on Combinatorial Mathematics and Computation Theory*, 181-187
- [5] G. Gutin, Characterizations of vertex pancyclic and pancyclic ordinary complete multipartite digraphs, *Discrete Mathematics*, 141 (1995) 153-162.
- [6] Junming Xu, *Theory and Application of Graphs*, Dordrecht, Boston and London, 2003.
- [7] J. Mitchem and E. Schmeichel, “Pancyclic and bipancyclic graphs—a survey,” *Graphs and Applications*, pp. 271–278, 1982.
- [8] Toru Araki, Edge-pancyclicity of recursive circulants, *Information Processing Letters* 88 (2003) 287-292.